

## 1. Logical URL in Servlets

### What is a Logical URL?

A **logical URL** is a **virtual URL pattern** that is mapped to a servlet, instead of using the actual physical file path.

It allows users to access servlets using meaningful and clean URLs

### Why use Logical URLs?

- Hides internal implementation details
- Makes URLs **user-friendly**
- Easier to manage and change mappings
- Improves security (no direct file access)

### How Logical URL works

Logical URLs are defined using:

- web.xml (deployment descriptor), or
- @WebServlet annotation

### Example using web.xml

```
<servlet>
  <servlet-name>MyServlet</servlet-name>
  <servlet-class>com.example.MyServlet</servlet-class>
</servlet>

<servlet-mapping>
  <servlet-name>MyServlet</servlet-name>
  <url-pattern>/login</url-pattern>
</servlet-mapping>
```

👉 Here:

- /login is the **logical URL**
- It maps to MyServlet

### Example using Annotation

```
@WebServlet("/login")
public class MyServlet extends HttpServlet {
    // servlet code
}
```

## Types of URL Patterns

1. **Exact Match** → /login
2. **Wildcard Path** → /user/\*
3. **Extension Mapping** → \*.do

## 2. ServletConfig Interface

### What is ServletConfig?

ServletConfig is an interface in the **Java Servlet API** that provides **configuration information** to a servlet.

Each servlet gets its own ServletConfig object.

### Purpose

- Pass initialization parameters to a servlet
- Provide access to servlet context
- Configure servlet behavior at deployment time

### How it is created

- Created by the **web container** (like Tomcat)
- Passed to servlet during initialization via init() method

### Example

#### web.xml configuration:

```
<servlet>
  <servlet-name>MyServlet</servlet-name>
  <servlet-class>com.example.MyServlet</servlet-class>

  <init-param>
    <param-name>username</param-name>
    <param-value>admin</param-value>
  </init-param>
</servlet>
```

#### Servlet code:

```
public class MyServlet extends HttpServlet {  
  
    public void init(ServletConfig config) {  
        String user = config.getInitParameter("username");  
        System.out.println(user);  
    }  
}
```

### **Methods of ServletConfig**

| <b>Method</b>                 | <b>Description</b>                         |
|-------------------------------|--------------------------------------------|
| getInitParameter(String name) | Returns value of a specific init parameter |
| getInitParameterNames()       | Returns all init parameter names           |
| getServletContext()           | Returns the ServletContext object          |
| getServletName()              | Returns the name of the servlet            |